National University of Computer and Emerging Sciences

Islamabad Campus

Probability and Statistics (MT2005)

BSCS(All Sections)

Course Instructors

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Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 75

Total Questions: 4

Date: April 09, 2025

Roll No ______ Section ______ Student Signature ______

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Attempt all the questions

CLO # 3: Define theory of random variable, illustrate the use of CDFs, PDFs and PMFs of continuous as well as discrete nature.

Question # 1 **[18 marks]**

(i) A probability density function is defined by $f(x)$ as follows:

$$f(x) = \begin{cases} \frac{x}{4}, & a \leq x \leq b \\ 0, & \text{elsewhere} \end{cases}$$

(a) If the median is 3, what are the lower and upper bounds of this distribution?

(b) Determine the score of the 85th percentile.

(ii) If a random variable X is defined such that:

$$E[(X-1)^2] = 10 \quad \text{and} \quad E[(X-2)^2] = 6,$$

find $E(X)$, $\text{Var}(X)$ and $\text{Var}(3X-2)$. (6)

(iii) Let X be a random variable with probability density function:

$$f(x) = \begin{cases} \frac{3}{2}(1-x^2), & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the cumulative distribution function $F(x)$. (3)

(b) Compute the mode of the distribution of X. (4)

(iv) A discrete random variable X has the following probability distribution:

X	-1	0	1
$P(X)$	0.2	0.4	0.4

Compute the skewness of X and hence determine the shape of the distribution. (5)

CLO # 4: Transform the given information into PMFs, PDFs, and CDFs, and express the probability of events based on statistical data.

Question # 2 **[27 marks]**

(i) Two discrete random variables X and Y have the following joint probability distribution:

$X \backslash Y$	1	2	3
0	0.1	0.2	0.1
1	0.2	0.1	0.3

(a) Compute $P(X = 1, Y = 2)$. (1)

(b) Find the marginal probability distribution of X and Y. (4)

(c) Are X and Y independent? (2)

(d) Find $E(X \mid Y = 2)$ and $\text{Var}(X \mid Y = 2)$. (5)

(ii) Let X and Y be two continuous random variables with the following joint probability density function (PDF):

$$f(x, y) = \begin{cases} c(x + y), & 0 \le x \le 2, \quad 0 \le y \le x, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the value of c to make this a valid joint PDF. (3)

(b) Compute $P(1 \le X \le 2, 0.5 \le Y \le 1)$. (3)

(c) Compute $P(Y \le 1 \mid X = 1.5)$. (5)

(iii) The processing time of an algorithm is modeled as a random variable with a mean of 8 seconds and a variance of 4 seconds^2. Using Chebyshevs Inequality, determine:

(a) The probability that the processing time deviates from the mean by at least 4 seconds. (2)

(b) The probability that the processing time is between 6 and 10 seconds. (2)

CLO # 4: Transform the given information into PMFs, PDFs, and CDFs, and express the probability of events based on statistical data.

Question # 3 **[10 marks]**

(i) Let X be a random variable representing the time taken by a software engineer to debug a program. The cumulant generating function about the origin is given by

$$K_X(t) = 2t$$

Calculate the mean and variance using the cumulant generating function. (2)

(ii) The PMF of the negative binomial distribution is:

$$P(X = x) = \binom{x-1}{k-1} p^k (1-p)^{x-k}, \quad x = k, k+1, k+2, \ldots$$

where:

- p is the probability of success on each trial,
- k is the number of successes,
- x is the number of trials to produce k successes.

Show that: (4)

$$\sum_{x=k}^{\infty} P(X = x) = 1$$

(iii) For the geometric probability distribution, prove that the moment generating function (MGF) is given by (4)

$$M_X(t) = \frac{pe^t}{1-(1-p)e^t}, \quad \text{for } t < -\ln(1-p).$$

CLO # 5: Apply probabilistic principles to solve real-world problems across both discrete and continuous domains.

Question # 4 **[20 marks]**

A machine learning model is used to classify emails as spam or not spam. Answer the following questions based on the models predictions.

(i) It is known that the machine learning model correctly classifies spam emails with a probability of 0.9. What is the probability that a string of 4 emails are processed without a correct spam classification before the model correctly classifies a spam email on the 5th email? What is the expected number of emails that are not classified as spam before the first successful spam classification? (4)

(ii) The model needs to correctly classify 4 spam emails. If the probability of success for each email is 0.9, what is the probability that the 4th successful classification occurs on the 7th processed email? (3)

(iii) A dataset contains 100 emails, of which 30 are spam and 70 are not spam. If a random sample of 10 emails is chosen for manual verification, what is the probability that exactly 4 of them are spam? (3)

(iv) The model assigns each email to one of three categories: spam (40%), promotional (35%), and regular (25%). If 25 emails are classified, what is the probability that exactly 10 are spam, 8 are promotional, and 7 are regular? (3)

(v) The system receives an average of 20 spam emails per hour. What is the probability that exactly 25 spam emails are received in the next hour? (3)

(vi) The model classifies each email independently, and the probability of correctly identifying spam is 90%. The model is tested on 7 emails. The model will be rejected if it classifies more than 2 emails incorrectly. What is the probability that the model will be rejected after processing the 7 emails? (4)